Continuity and the Intermediate Value Theorem

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Three-Part Definition of Continuity

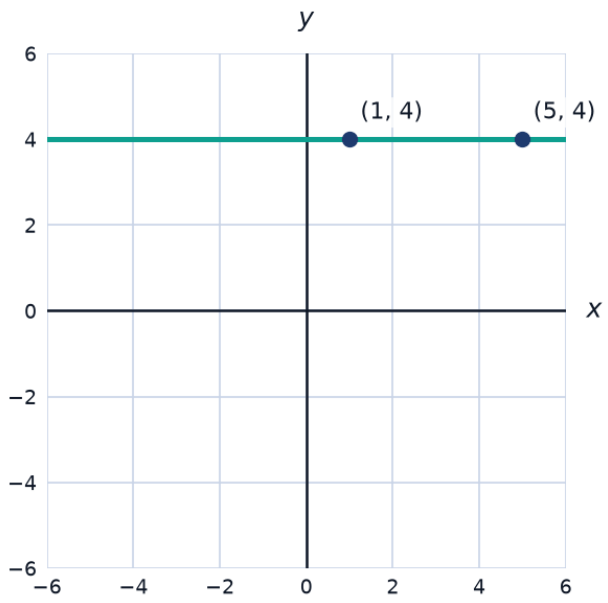
A rectangular field is to be enclosed by a fence. One side of the field is adjacent to a river and does not require fencing. The farmer has 120 meters of fencing material. Let the dimensions of the field be x meters perpendicular to the river and y meters parallel to the river. The area of the field is given by $A = xy$. Part A: Express the area A as a function of x. Part B: Determine the dimensions x and y that maximize the area of the field. Provide the maximum possible area.

2 [Foundational] • Target: Piecewise Functions & Continuity

The function $f(x)$ is defined piecewise as: $f(x) = \{ax^2 + 1, , x < 2; 5x - b, , x \geq 2$ If $f(x)$ is continuous at $x = 2$, and the limit of $f(x)$ as $x \to 2$ is 9, find the values of a and b.

3 [Foundational] • Target: Intermediate Value Theorem (IVT)

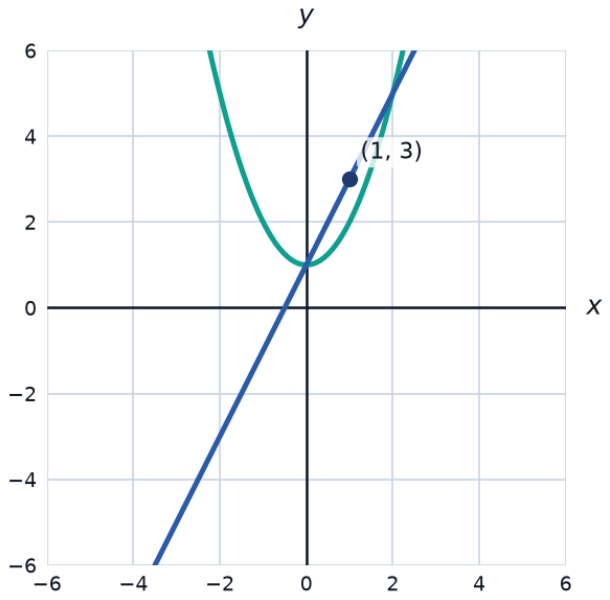
The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.



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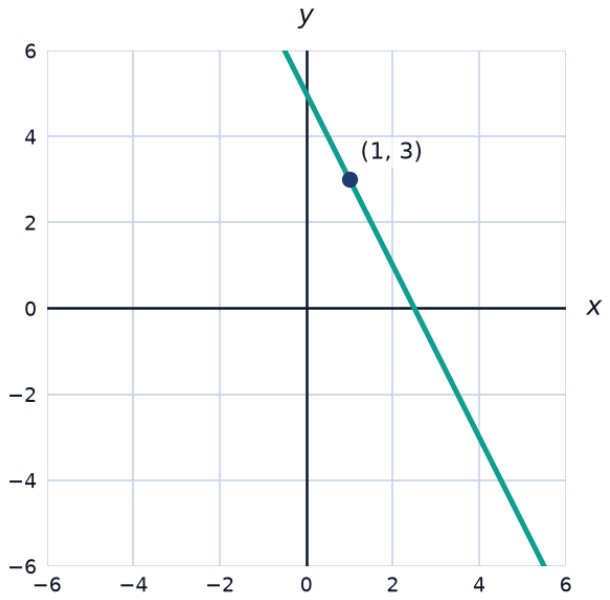
4 [Foundational] • Target: Extreme Value Theorem (EVT)

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.



5 [Foundational] • Target: Three-Part Definition of Continuity

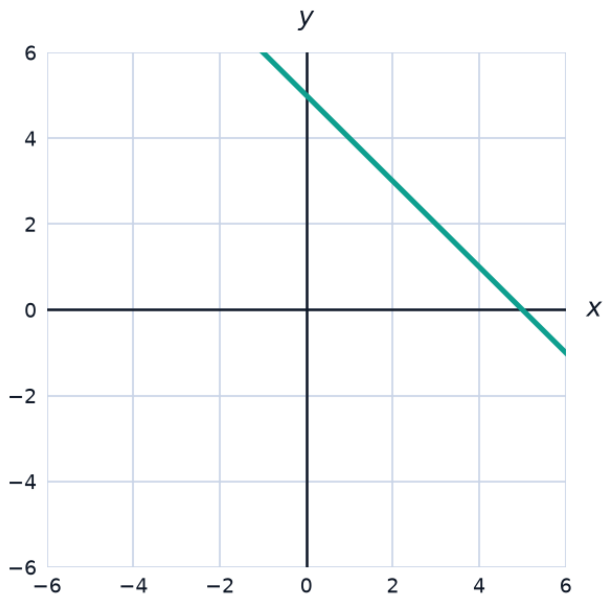
Consider the differential equation $\frac{dy}{dx} = kx - 3$, where k is a constant. The graph of a particular solution $y = f(x)$ to this differential equation is shown below. The tangent line to the graph at $x = 1$ has the equation $y = -2x + 5$. Find the value of k.



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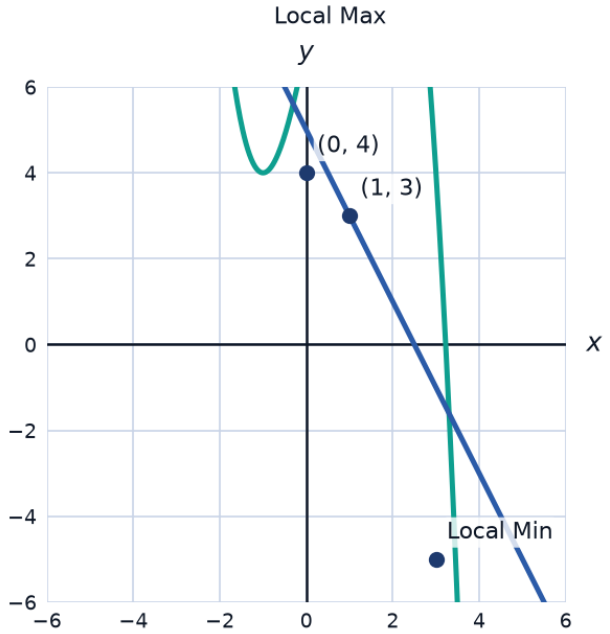
6 [Foundational] • Target: Piecewise Functions & Continuity

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?



7 [Foundational] • Target: Intermediate Value Theorem (IVT)

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = -1$ and a local minimum at $x = 3$. The tangent line to the graph of f at $x = 1$ has the equation $y = -2x + 5$. If $f(0) = 4$ and $f(3) = -5$, find the value of $f'(1)$.

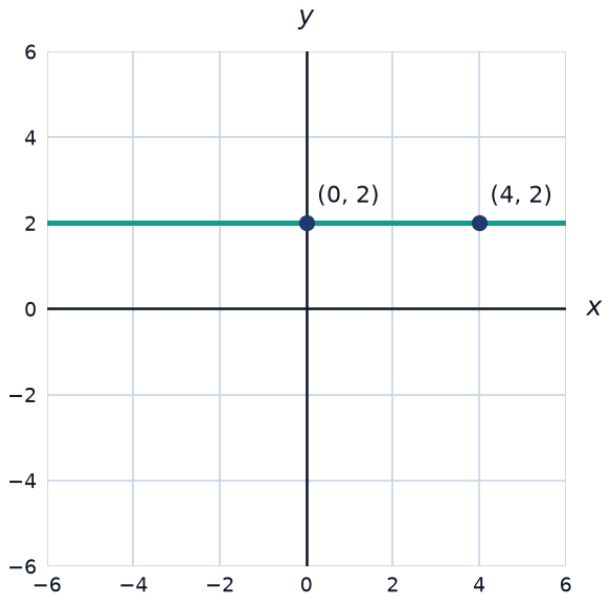


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

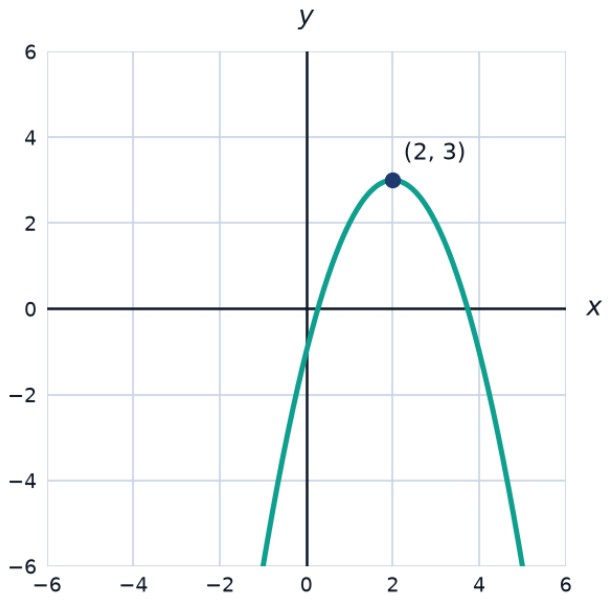
8 **[Intermediate]** • **Target: Extreme Value Theorem (EVT)**

The graph of a function $f(x)$ is shown above. What is the instantaneous rate of change of $f(x)$ at $x = 2$?



9 **[Intermediate]** • **Target: Three-Part Definition of Continuity**

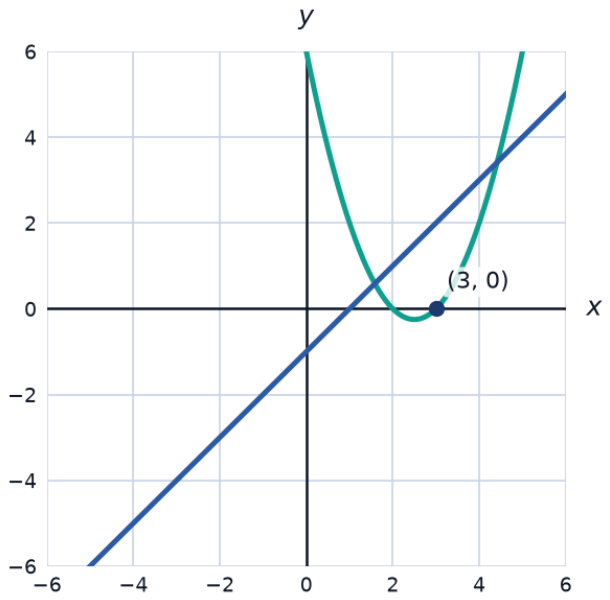
The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.



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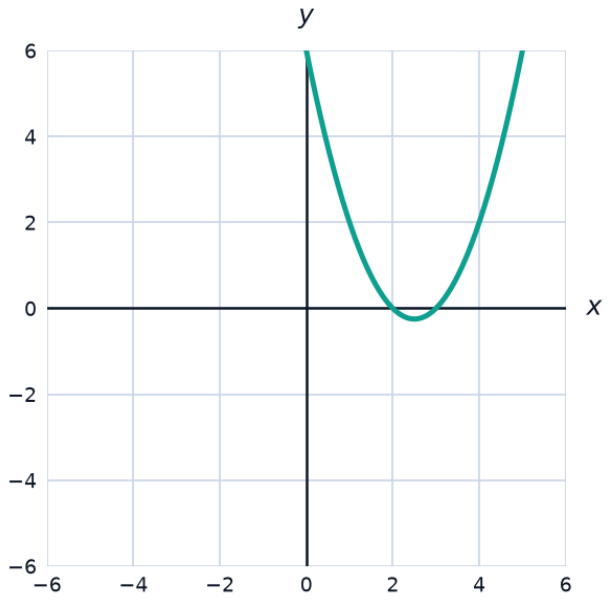
10 **[Intermediate]** • **Target: Piecewise Functions & Continuity**

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $\lim_{x \to 3} f(x)$?



11 **[Intermediate]** • **Target: Intermediate Value Theorem (IVT)**

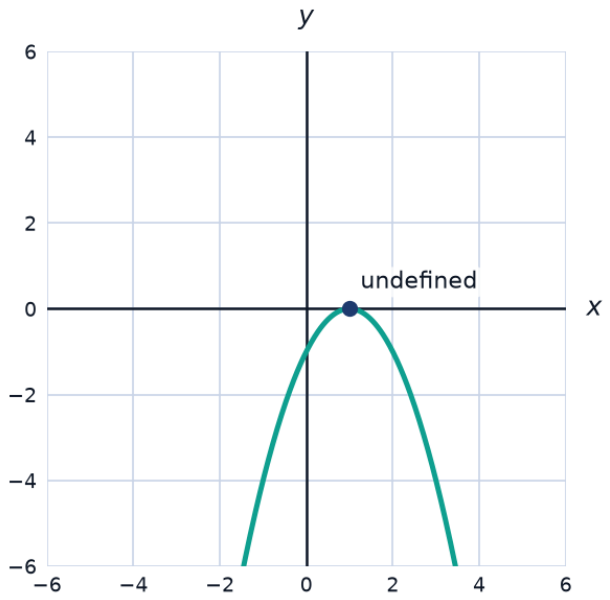
Consider the function $f(x) = x^2 - 5x + 6$. What is the limit of $f(x)$ as x approaches 2?



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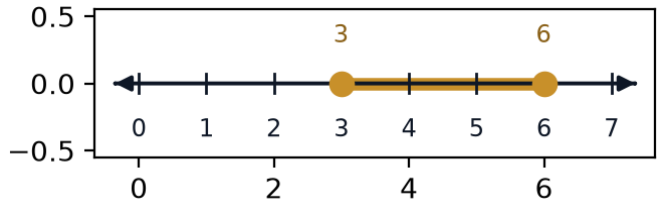
12 **[Intermediate]** • **Target: Extreme Value Theorem (EVT)**

Consider the function $f(x)$ graphed below. What is the limit of $f(x)$ as x approaches 1, i.e., $\lim_{x \to 1} f(x)$?



13 **[Intermediate]** • **Target: Three-Part Definition of Continuity**

Consider the function $f(x) = 4$. What is the value of the definite integral $\int_3^6 f(x)\,dx$?



14 **[Intermediate]** • **Target: Piecewise Functions & Continuity**

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Intermediate Value Theorem (IVT)** • **Cross-Lesson: 2.01 The Definition of the Derivative**

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

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16 [Challenging] • Target: Extreme Value Theorem (EVT) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Consider the function $f(x)$ defined by $f(x) = \frac{x^2 - 9}{x - 3}$ for $xeq3$. Let $g(x)$ be a function such that $g(x) = \{f(x), \text{if } x < c; k, \text{if } x \geq c$. If $g(x)$ is continuous at $x = c$ and the limit of $g(x)$ as x approaches c from the left is 5, determine the value of k.

17 [Challenging] • Target: Three-Part Definition of Continuity • Cross-Lesson: 3.01 The Chain Rule

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.

18 [Challenging] • Target: Piecewise Functions & Continuity • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Consider the function $y = g(x)$ and its inverse $x = h(y)$. The tangent to the graph of $g(x)$ at $x = a$ has a gradient of $\frac{1}{3}$. If $g(a) = 5$, what is the gradient of the tangent to the graph of $h(y)$ at $y = 5$?

19 [Challenging] • Target: Intermediate Value Theorem (IVT) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

20 [Challenging] • Target: Extreme Value Theorem (EVT) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A differentiable function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = 3x - 2$. If $f(1) = 5$, what is the value of $f(3)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Three-Part Definition of Continuity

Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $f(c)$ is continuous on (a, b), $f(c) = 0$, and $f'(c) \neq 0$

d. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

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22 [Foundational] • Target: Three-Part Definition of Continuity

Let $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. What value must be assigned to $f(3)$ to make f continuous at $x = 3$?

a. 3

b. 6

c. 0

d. No value can make it continuous

23 [Foundational] • Target: Three-Part Definition of Continuity

The function $g(x) = \frac{1}{x - 2}$ has what type of discontinuity at $x = 2$?

a. Infinite (essential) discontinuity

b. Removable discontinuity

c. Jump discontinuity

d. Oscillating discontinuity

24 [Foundational] • Target: Piecewise Functions & Continuity

For what value of k is $f(x) = kx^2 + 1$ for $x \leq 2$, and $f(x) = 3x - 1$ for $x > 2$ continuous at $x = 2$?

a. 1

b. 2

c. 3

d. 0

25 [Foundational] • Target: Piecewise Functions & Continuity

If $g(x) = \frac{\sin(ax)}{x}$ for $x \neq 0$, and $g(0) = 4$, what value of a ensures continuity at $x = 0$?

a. 1

b. 2

c. 4

d. $\frac{1}{4}$

26 [Foundational] • Target: Piecewise Functions & Continuity

The function $h(x) = x + 2$ for $x < 1$, and $h(x) = 2x + 1$ for $x \geq 1$ has what property at $x = 1$?

a. Continuous because both one-sided limits equal 3

b. Jump discontinuity of size 1

c. Infinite discontinuity

d. Undefined at $x = 1$

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27 [Foundational] • Target: Intermediate Value Theorem (IVT)

What are the hypotheses of the Intermediate Value Theorem for a function f on interval $[a, b]$?

a. f must be continuous on the closed interval $[a, b]$

b. f must be differentiable on (a, b)

c. $f(a)$ and $f(b)$ must have opposite signs

d. f must be strictly increasing on $[a, b]$

28 [Foundational] • Target: Intermediate Value Theorem (IVT)

Let f be continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 7$. Does the IVT guarantee a solution to $f(c) = 0$ for some $c \in (1, 5)$?

a. Yes, because 0 lies between -3 and 7

b. No, f must also be differentiable

c. No, f must be a polynomial

d. Only if $f'(x) > 0$ for all x

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Intermediate Value Theorem (IVT)

On which interval is the polynomial $P(x) = x^3 - 3x - 1$ guaranteed to have a real zero by the IVT?

a. $[1, 2]$

b. $[0, 1]$

c. $[-1, 0]$

d. $[2, 3]$

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30 **[Intermediate]** • **Target: Extreme Value Theorem (EVT)**

What does the Extreme Value Theorem guarantee for a function f continuous on a closed interval $[a, b]$?

a. f attains both an absolute maximum and an absolute minimum on $[a, b]$

b. f has a horizontal tangent on (a, b)

c. $f'(c) = 0$ for some $c \in (a, b)$

d. f is invertible on $[a, b]$

31 **[Intermediate]** • **Target: Extreme Value Theorem (EVT)**

Why does the function $f(x) = \frac{1}{x}$ NOT attain an absolute maximum on $(0, 2]$?

a. The interval is open at 0 and f increases without bound as $x \to 0^+$

b. f is not differentiable on $(0, 2]$

c. f is bounded on $(0, 2]$

d. $f(2) = 0.5$ is the maximum

32 **[Intermediate]** • **Target: Extreme Value Theorem (EVT)**

Find the absolute maximum value of $f(x) = -x^2 + 4x + 1$ on the closed interval $[0, 3]$.

a. 5

b. 4

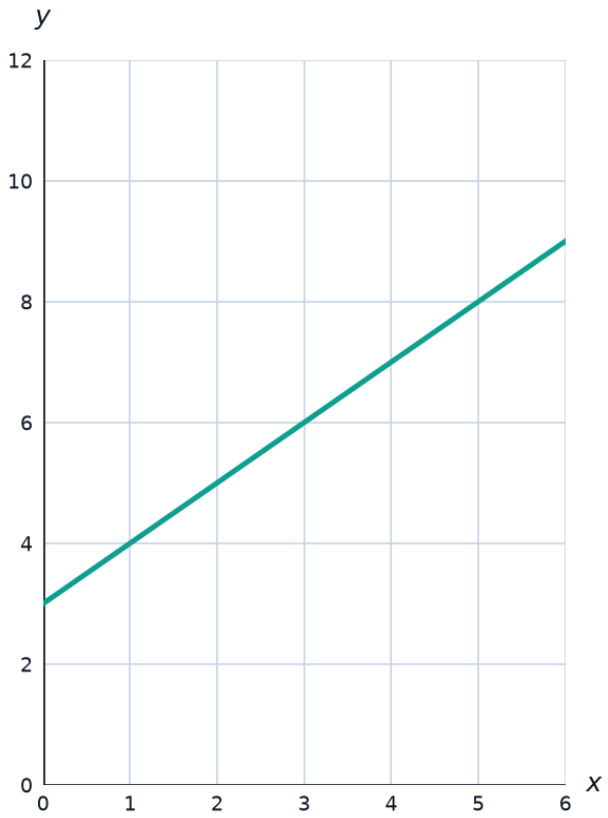
c. 1

d. 6

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33 **[Intermediate]** • **Target: Three-Part Definition of Continuity**

Consider the function $f(x)$ defined piecewise as: $f(x) = \begin{cases} \frac{x^2-9}{x-3}, \text{ if } xeq3; \\ k, \text{ if } x = 3 \end{cases}$ For the function $f(x)$ to be continuous at $x = 3$, the value of k must be such that the limit of $f(x)$ as x approaches 3 exists and is equal to $f(3)$. Which of the following statements is true regarding the continuity of $f(x)$ at $x = 3$ if k is set to satisfy this condition, and what is the value of that limit?



a. The function is continuous at $x = 3$ and the limit is 6.

b. The function is continuous at $x = 3$ and the limit is 9.

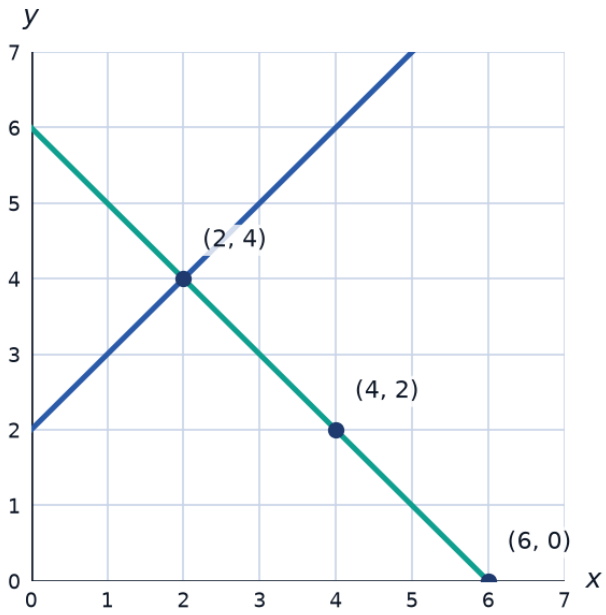
c. The function is not continuous at $x = 3$ regardless of the value of k.

d. The function is continuous at $x = 3$ and the limit is 3.

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34 **[Intermediate]** • **Target: Piecewise Functions & Continuity**

The graph of the derivative of a function $f(x)$ is shown below. Given that $f(2) = 5$, determine the value of $f(6)$.



a. 13

b. 18

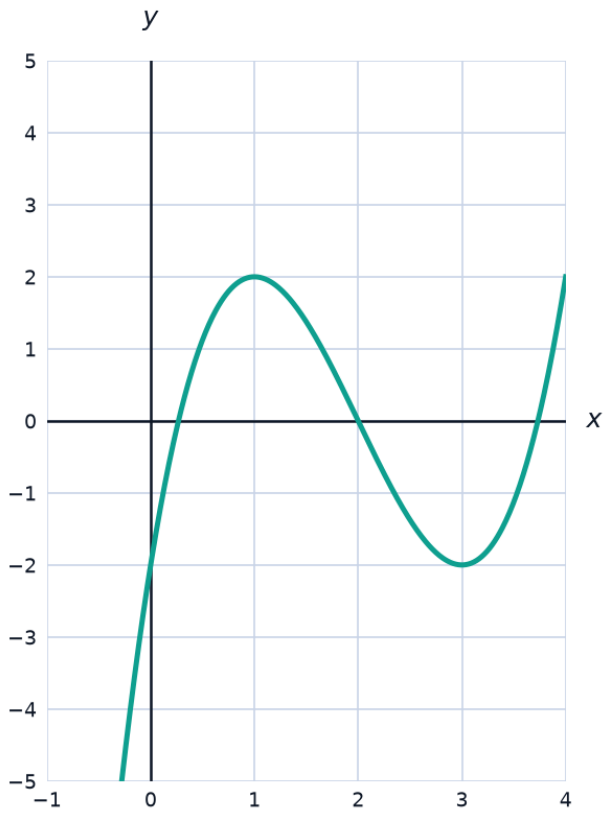
c. 23

d. 28

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35 **[Intermediate]** • **Target: Intermediate Value Theorem (IVT)**

Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.



a. -6

b. -12

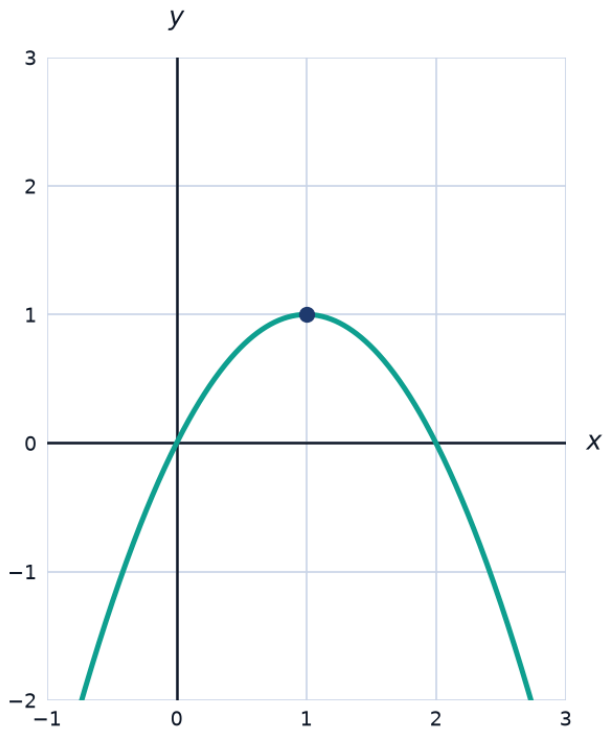
c. 0

d. 6

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36 [Intermediate] • Target: Extreme Value Theorem (EVT)

The graph of a differentiable function $f(x)$ is shown below. Which of the following statements about the derivative of $f(x)$, denoted $f'(x)$, at $x = 1$ is true?



a. $f'(1)$ is positive.

b. $f'(1)$ is negative.

c. $f'(1)$ is zero.

d. $f'(1)$ cannot be determined from the graph.

37 [Intermediate] • Target: Three-Part Definition of Continuity

The graph of a function $g(x)$ is shown above. The function $g(x)$ is defined by $g(x) = \frac{1}{3}x^3 - ax^2 + 2x + 1$, where a is a positive constant. The graph has a tangent line at $x = 1$ which is parallel to the line $y = -2x + 5$. Find the value of the constant a.

a. 1

b. 2

c. 3

d. 4

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Piecewise Functions & Continuity • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Let $f(x)$ be a function such that $f'(x) = \cos(x^2)$ and $g(x)$ be a function defined by $g(x) = \sqrt{x}$. ; ; \textbf{(a)} Find the derivative of the composite function $h(x) = f(g(x))$. ; ; \textbf{(b)} If $f(1) = \pi$, find the value of $h'(4)$.

a. $\frac{\cos(4)}{2}$

b. $\frac{\pi\cos(4)}{2}$

c. $\frac{\cos(16)}{2}$

d. $\frac{\pi\cos(16)}{2}$

39 [Challenging] • Target: Intermediate Value Theorem (IVT) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

What is the value of $\lim_{h\to 0} \frac{(2+h)^4 - 16}{h}$?

a. 8

b. 16

c. 32

d. 64

40 [Challenging] • Target: Extreme Value Theorem (EVT) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x) = x^2 - 3x$. If $g(x)$ is a function such that $h(x) = f(g(x))$, and at $x = 2$, $g(2) = 5$ and $g'(2) = 3$, find $h'(2)$.

a. 21

b. 30

c. 15

d. 6

41 [Challenging] • Target: Three-Part Definition of Continuity • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Let $f(x) = \sin(x^2)$. What is the derivative of $f(x)$?

a. $\cos(x^2)$

b. $2x\cos(x^2)$

c. $\sin(2x)$

d. $-2x\cos(x^2)$

42 [Challenging] • Target: Piecewise Functions & Continuity • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

If $f(x) = \sin(x^2)$, then $f'(x) =$

a. $\cos(x^2)$

b. $2x\cos(x^2)$

c. $2\cos(x^2)$

d. $x\cos(x^2)$

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43 [Challenging] • Target: Intermediate Value Theorem (IVT) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

$\frac{d}{dx}(\cos x) =$

a. $\sin x$

b. $-\sin x$

c. $\sec^2 x$

d. $-\cos x$

44 [Challenging] • Target: Extreme Value Theorem (EVT) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

If $f(x) = x^4$, find $f'(3)$.

a. 108

b. -108

c. 104

d. 112

45 [Challenging] • Target: Three-Part Definition of Continuity • Cross-Lesson: 7.01 Differential Equations and Slope Fields

If $f(x) = x^4$, find $f'(-1)$.

a. -4

b. 4

c. 0

d. -8

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